INCH-POUND

MIL-DTL-83196C(USAF)
30 November 1998
SUPERSEDING
MIL-U-83196B(USAF)
26 November 1986

DETAIL SPECIFICATION

UNDERWEAR, DOUBLE-KNITTED, ANTI EXPOSURE CWU-22/P

Inactive for new design after 1 December 1998

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of extreme cold weather underwear, designated CWU-22/P.

1.2 Classification. The underwear should be in the following sizes as specified (see 6.2). The civilian equivalent sizes are in parentheses.

Size 1 (32-33)	Size 5 (40-41)
Size 2 (34-35)	Size 6 (42-43)
Size 3 (36-37)	Size 7 (44-45)
Size 4 (38-39)	Size 8 (46-47)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. These lists do not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet the requirements specified in the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Resources & Logistics Services Division, SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Kelly AFB, Texas 78241-6425, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8475

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of these documents are those listed in the specific issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50155	-	Cloth, Muslin, Cotton
A-A-55621	-	Fasteners, Snap, Style 3 Pronged Ring Head Type (16 LIGNE)
V-T-285	-	Thread, Polyester
V-T-295	-	Thread, Nylon
JJ-W-155	-	Webbing, Textile (Cotton Elastic)
CCC-C-432	-	Cloth, Sheeting, Cotton (Unbleached, Bleached, Dyed)
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)

DEPARTMENT OF DEFENSE

MIL-C-3735	-	Cuffs, Knit, Wrist and Ankle, and Cloth, Knitted
MIL-F-10884	-	Fastener, Snap

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, Pennsylvania 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues are those cited in the solicitation.

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DRAWINGS

AIR FORCE

68J236	-	Patterns - Underwear, Double-Knitted, Anti-Exposure CWU-22/P, Size 1
68J237	-	Patterns - Underwear, Double-Knitted, Anti-Exposure CWU-22/P, Size 2
68J238	-	Patterns - Underwear, Double-Knitted, Anti-Exposure CWU-22/P, Size 3
68J239	-	Patterns - Underwear, Double-Knitted, Anti-Exposure CWU-22/P, Size 4
71387	-	Patterns - Underwear, Double-Knitted, Anti-Exposure CWU-22/P, Size 5
71388	-	Patterns - Underwear, Double-Knitted, Anti-Exposure CWU-22/P, Size 6
71389	-	Patterns - Underwear, Double-Knitted, Anti-Exposure CWU-22/P, Size 7
71390	-	Patterns - Underwear, Double-Knitted, Anti-Exposure CWU-22/P, Size 8

(Application for copies should be addressed to SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Kelly AFB, Texas, 78241-6425.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of the documents which have been adopted by the DoD are those listed in the specific issue of the DoDISS cited in the solicitation. Unless otherwise specified, the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z1.4	-	Sampling Procedures And Tables For Inspection By Attributes (DoD adopted)

(Application for copies should be addressed to American Society for Quality Control, 611 East Wisconsin Avenue, P.O. Box 3005, Milwaukee, Wisconsin, 53201-3005 or to the American National Standards Institute, 11 West 42nd Street, New York, New York, 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 177	-	Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus (DoD adopted)
ASTM D 1777	-	Standard Test Method for Thickness of Textile Materials
ASTM D 3776	-	Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM D 5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA, 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials.

3.2.1 Basic fabric. The basic integral fabric shall comprise of a layer of oriented, high loft, continuous filament nylon yarns (nylon filler) sandwiched between linked layers of knitted cotton (see figure 1). The color of the nylon fibers shall be white or natural. The color of the knitted cotton shall be sage green and approximate USAF color shade No. 1509 (see 6.3). The nylon yarns shall be introduced evenly and in sufficient numbers between the tie stitches of the two layers of knitted cotton in such a manner that the length is in the direction of the greatest stretch and to meet the required thickness over the full width of the fabric. The finished fabric shall present a smooth (unpuckered) surface on both sides and shall conform to the requirements specified in table I.

TABLE I. Basic fabric.

Characteristics	Requirement
Total weight, ounces per square yard	
Minimum	20
Maximum	23
Knited cotton weight, ounces per square yard	
Minimum	11
Maximum	12
Nylon filler weight, ounces per square yard	
Minimum	9
Maximum	11
Thermal conductivity when compressed 25%	0.0474 to 0.0478
Thickness	
Uncompressed, inch	0.25 ± 0.01
Compressed (percent of uncompressed), minimum	80
Elongation, percent 1/	
Wales	35
Courses	70
Air permeability, cubic feet per square foot per minute, minimum	100
Width, inches	32 ± 1

1/ Minimum stretch at 1/3 maximum tensile load.

3.2.1.1 Nylon yarns. The nylon filler shall be delustered nylon yarns. The nylon yarns before crimping or texturing shall be 2 ply, 80 to 180 denier with a maximum filament size of 7 denier per filament. The nylon yarn shall be one of the following types:

a. A yarn with a pronounced crimp pattern of 4/64 ± 1/64 inch in height and 14 to 16 crimps per inch.

b. A yarn, having a minimum elongation of 300%, in a random crimp pattern, a fiber entanglement pattern, or a combination random crimp and fiber entanglement pattern.

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3.2.1.2 Knitted cotton and tie threads. The knitted cotton and the tie threads shall be made of evenly carded and drawn 2-ply yarns with sufficient twist for the requirement. The tie threads shall be an integral part of the cotton knit fabric.

3.2.2 Elastic webbing. The elastic webbing for the waistband shall conform to Type I, Class 14 of JJ-W-155. The elastic webbing shall be 2 inches wide. The color of the elastic webbing shall approximately match the color of the basic fabric. The elastic webbing shall be colorfast to laundering and perspiration.

3.2.3 Thread. The thread for sewing the underwear shall conform to Type I, Class 1 or 3, subclass A, letter size A or B of V-T-285, or to Type I, Class 1, letter size A of B of V-T-295. The color of the thread shall be sage green and approximate USAF color shade No. 1509 (see 6.3). The thread shall be colorfast to laundering and perspiration.

3.2.4 Snap fasteners. The snap fasteners shall be pronged-ring head type conforming to A-A-55621-1B, -3B, -4B, and -5B; style 3 of MIL-F-10884.

3.2.5 Fly facing. The fabric for the fly facing shall conform to Type I or II of A-A-50155; or to Type I or II, Class 3 of CCC-C-432. The fabric for the fly facing shall be sage green and match USAF color shade No. 1509 (see 6.3). The fly facing fabric shall be colorfast to laundering and perspiration.

3.2.6 Collar, wristlets, and anklets. The fabric for the collar, wristlets, and anklets shall conform to MIL-C-3735, except that the wool requirements, test methods, and shrink requirements shall not apply. The fabric for the collar shall be Type I, Class 3 of MIL-C-3735. The fabric for the wristlets shall be Type I, Class 2 of MIL-C-3735. The fabric for the anklets shall be Type I, Class 1 of MIL-C-3735. The fabric shall be made of alternating cotton and nylon yarns. The cotton yarns shall be evenly carded, drawn, and spun into singles, with sufficient twist to meet the requirements of MIL-C-3735. The nylon yarns shall be spun on the worsted system and plied into 2-ply yarn with sufficient twist to meet the requirements of MIL-C-3735. For the anklets and wristlets, three courses of single, cotton yarn shall alternate with one course of 2-ply spun nylon yarn.

3.2.7 The anklets shall be finished to a folded width of 3 ± 1/4 inches and an open length of 9 ± 1/4 inches. The wristlets shall be finished to a folded width of 2-1/2 ± 1/4 inches and an open length of 9 ± 1/4 inches. The weight of the anklets shall be a minimum of 12 ounces per dozen pairs. The weight of the wristlets shall be a minimum of 9 ounces per dozen pairs. The fabric for the collar, wristlets, and anklets shall be colorfast to laundering and perspiration.

3.3 Design. The underwear shall consist of two pieces: a pullover type undershirt and a conventional fly opening underpants. The underwear shall have knitted wristlets, anklets, and crew neck type collar. The underpants shall have an elastic waistband and a reinforced fly section. Figure 3 shows a general design of the underwear and is furnished for information only.

3.4 Construction. The underwear shall be constructed as specified in table II, using sewn seams; however, the manufacturer is not required to follow the exact sequence of operations as listed therein. The underwear shall be cut in accordance with the patterns specified in 3.5.

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TABLE II. Sewing operations for underpants.

Sequence Number	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches per Inch
1	Overedge stitch underpants. Overedge stitch all raw edges of the basic two parts of the underpants and the crotch piece.	504 or 505	EFd-1	20-24
2	Join underpants. a. Join the side darts. b. Join the back seam. c. Join the crotch piece to the right and left backs. d. Join the facing to the right and left fly opening using seam type SSae-1 on the top and front edges and seam type LSd-1 on the back edges.	 606 606 606 301 or 304	 FSa-1 FSa-1 FSa-1 SSae-1 LSd-1	 12-14 12-14 12-14 8-10 8-10
3	Install fly. Place the left fly over the right fly with the edge of the left fly even with the inside row of stitching on the right fly. Stitch the left fly to the right fly at the bottom area with one row of stitching. The stitching shall start at the point that the outer edge of the left fly is joined to the crotch seam. The stitching shall continue on a diagonal line upward across the left fly to the inside row of stitching on the fly at a point approximately 1-1/2 inches from the crotch seam (see figure 2).	301 or 304	SSv-1	8-10
4	Cut and attach waistband. a. Cut the elastic webbing for the waistband the required length. The length in inches for sizes 1, 2, 3, 4, 5, 6, 7, and 8 shall be 22, 24, 26, 28, 30, 32, 34, and 36 (±1/4 inch), respectively. b. Stitch the elastic webbing to the top of the underpants along each end and across the bottom edge with the elastic webbing overlapping the underpants 1/4 ± 1/8 inch. The elastic webbing shall be equally distributed around the circumference of the underpants.	 606 or 607	 FSa-1	 12-14
5	Install snap fasteners. Install the four snap fasteners, the pronged rings in the left fly, and studs in the right (under) fly. The top snap fastener shall be 3/4 inch from the top edge. The second snap fastener shall be 2-1/4 inches from the top edge. The bottom snap fastener shall be 3 inches from the bottom of the fly. The third snap fastener shall be equidistant from the top and the bottom snap fasteners.			

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TABLE II. Sewing operations for underpants. - Continued

Sequence Number	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches per Inch
6	Overedge stitch and join anklet.			
	a. Overedge stitch the raw edges of the knitted anklets catching both edges of the knit.	504 or 505	EFd-1	20-24
	b. Join the anklets to the leg terminals of the underpants.	301 or 504	SSa-1	12-14
	c. Cover stitch the raw seam edges.	402, 406, or 606	SSh-2(a)	20-24
7	Attach label to underpants. Center the label over the back seam 2-1/2 ± 1/4 inches down from the finished top of the underpants. Join the label to the underpants with a single row of stitching on all four sides.	301 or 304	LSbj-1	8-10
8	Trim thread ends. Trim all thread ends and remove loose threads.			
9	Cut undershirt. Cut the back, front, sleeves, and knit collar.			
10	Overedge stitch undershirt. Overedge stitch all raw edges of the back, front, and sleeves of the undershirt, including the underarm vent cutouts.	504 or 505	EFd-1	8-10
11	Join undershirt.			
	a. Join the front to the back of the shoulders.	606	FSa-1	12-14
	b. Join the sleeves to the armholes.	606	FSa-1	12-14
	c. Join the side seams.	606	FSa-1	12-14
	d. Join the wristlets to the sleeve ends.	606	FSa-1	12-14
	e. Join the ends of the knit collar with a single row of stitching 1/2 inch from the edge.	301	SSa-1	8-10
12	Overedge stitch and join collar to undershirt.			
	a. Fold the collar lengthwise with the raw edges of the seam to the inside and overedge stitch the two edges together.	504 or 505	EFd-1	20-24
	b. Join the collar to the neck opening, joining seam for the collar to be the back of the undershirt.	606	FSa-1	12-14
13	Attach label to undershirt. The label shall be positioned at the center of the back 1 ± 1/4 inch down from the collarline and stitched to the undershirt with a continuous row of stitching on all four sides.	301 or 304	LSbj-1	8-10

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3.4.1 Stitches, seams, and stitchings. The stitches, seams, and stitchings specified in table II shall be in accordance with FED-STD-751. The overedge stitching shall extend not less than 3/16 inch from the raw edge. The overedge machining shall not be equipped with a knife, or the knife shall be removed. The rows of stitching on each side of seam Type FSa-1 shall be a minimum of 3/32 inch and a maximum of 5/32 inch from the center of the seamline. Stitches per inch shall be as specified in table II.

3.4.2 Finished measurements (underpants). The finished measurements of the underpants shall conform to measurements in table III. With the underpants laid out flat and face up, under no tension, the finished measurements shall be taken as follows:

a. Outseam - from the top edge of the waistband to the top of the knit anklet.

b. Inseam - down the inseam from the center of the crotch to the top of the knit anklet.

c. Waist - across the top edge of the waistband from folded edge to folded edge.

d. Seat - with the fly closed, 2 inches above the crotch from folded edge to folded edge.

TABLE III. Finished measurements for the underpants.

Size of Underpants	Finished Measurements in Inches			
	Outseam (+1, -3/4)	Inseam (+1, -3/4)	Waist (+1, -3/4)	Seat (+1, -3/4)
1	36	26	12	19-3/4
2	37	26-1/2	13	20-3/4
3	38	27-3/4	14	21-3/4
4	39	28-1/2	15	22-3/4
5	40	29-1/2	16	23-3/4
6	41	30-1/2	17	24-3/4
7	42	31	18	25-3/4
8	43	32	19	26-3/4

3.4.3 Finished measurements (undershirt). The finished measurements of the undershirt shall conform to table IV. With the undershirt laid out flat and face up, in a relaxed condition, with the side seams on the folded edge and the sleeves extended horizontally perpendicular to the body, the finished measurements shall be taken as follows:

a. Chest - below the underarm vents and across the front from the folded edge to the folded edge.

b. Back length - at the center of the back from the collar seam to the bottom of the undershirt.

c. Sleeve length - along the top folded edge of the sleeve from the intersection of the armhole seam and the shoulder seam to the top of the wristlet.

d. Shoulder seam - along the shoulder seam from the intersection of the shoulder seam and the collar seam to the intersection of the shoulder seam and the armhole seam.

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TABLE IV. Finished measurements for the undershirt.

Size of Undershirt	Finished Measurements in Inches			
	Chest (±1/2)	Back Length (±1/2)	Sleeve Length (±1/2)	Shoulder Seam (±1/4)
1	15-1/2	24-1/4	19-1/4	4-1/4
2	16-1/2	24-1/2	20	4-1/2
3	17-1/2	24-3/4	20-3/4	4-3/4
4	18-1/2	25	20-1/2	5
5	19-1/2	25-1/4	22-1/4	5-1/4
6	20-1/2	25-1/2	23	5-1/2
7	21-1/2	25-3/4	23-3/4	5-3/4
8	22-1/2	26	24-1/2	6

3.5 Patterns. The underwear shall conform to the patterns referenced on Drawings 68J236, 68J237, 68J238, 68J239, 71387, 71388, 71389, or 71390, as applicable. Standard patterns will be furnished by the Government (see 6.2). The patterns, which are intended for a butted seam construction, have no seam allowance. The standard patterns are intended to be used only as a guide for making the manufacturer's working patterns. The patterns shall not be altered without prior approval of the procuring activity.

3.6 Label. Each of the undershirts and underpants shall have a label placed at the center back as specified in table II. The label shall conform to Type I, Classes 1 and 2 of DDD-L-20 and shall be colorfast when laundered. The label shall be approximately 1-3/4 by 2-3/4 inches. The manufacturer shall include the applicable information on the label.

Underwear, Double-Knitted, Anti Exposure CWU-22/P
Specification MIL-DTL-83196C(USAF)
Manufacturer's Identification
Contract No.
Date of Manufacture (Month and Year)
Size (include civilian equivalent size from 1.2)
Stock No.

3.7 Workmanship. The underwear shall be constructed in accordance with commonly accepted industrial workmanship standards. The finished underwear shall be clean and free from any defects that might adversely affect appearance or serviceability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When required (see 6.2), first article inspection shall consist of the examinations in 4.4 and 4.5 and the test in 4.6.

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4.2.1 First article test sample. First article test samples shall consist of 1-1/2 yards of the basic fabric, four undershirts, and four underpants. One each of the undershirts and underpants shall be furnished in sizes 1, 3, 5, and 7. The samples shall be representative of the construction, workmanship, components, and material used during production.

4.3 Conformance inspection. Conformance inspection shall include the examinations in 4.4 and 4.5 and the test in 4.6. The inspection levels shall be in accordance with ANSI/ASQC Z1.4, level II for the examination in 4.4 and level S-3 for the examination in 4.5.

4.4 Examination of end item. The end item (undershirt and underpants) shall be examined for defects as specified in table V. The end item shall be inspected for compliance with the material, design, and construction requirements specified in 3.2, 3.3, and 3.4.

TABLE V. Classification of defects.

Defect	Major	Minor
Defects and workmanship damages 1/		
a. Any weakening defect such as a hole, run in knitted fabric, loose slub, or smash that might develop into a hole		
b. Scissors or knife cut, tear, mend, burn, or needle chew that could develop into a hole		
Badly shaded parts		X
Cleanliness		
a. A permanent spot or stain		X
b. Removable spot or stain, clearly noticeable		X
c. Thread ends not trimmed or loose threads not removed		X
Components and assembly		
a. Any defective component	X	
b. Any component part omitted	X	
c. Any required operation omitted or improperly performed	X	
Cutting		
Any component part not cut in accordance with specified requirements	X	
Seams and stitchings 2/		
a. Seam twisted, puckered, or pleated	X	
b. Part of garment caught in any unrelated operation or stitching	X	
c. Thread breaks or ends of stitching not caught in any other seams or stitching		X
d. Gage of stitching incorrect, resulting in a faulty or open seam		X
e. Gage of stitching irregular or excessively uneven stitching that affects appearance		X
f. Any open seam 1/4 to 3/4 inch 3/		X

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TABLE V. Classification of defects. - Continued

Defect	Major	Minor
g. Any open seam over 3/4 inch	X	
h. Stitches skipped over 1/4 inch		X
i. Omission of overedge stitching on raw edges (seaming) more than 1/2 inch		X
j. Runoffs on joining seams that result in open seams 3/		
k. Not specified seam or stitch type		X
l. Stitch tension loose resulting in a loose seam	X	
m. Stitches per inch more or less than 3 stitches per inch than specified		X
Label		
a. Label missing, incorrect, or illegible	X	
b. Label not securely stitched on all sides		X
c. Label not correctly positioned		X
Snap fasteners		
a. One or more sockets or studs broken, bent, or otherwise defective, not functioning properly	X	
b. Snap fasteners not properly clinched, too loosely or tightly clinched, which cuts surrounding fabric	X	
c. Sockets and studs not aligned properly resulting in a bulge or twist on fly parts		X

1/ Any material defects or workmanship damage that is inconspicuous and does not weaken the material shall be classified as a minor defect. Material defects or workmanship damage that weakens the material, or does not weaken the material but is conspicuous when the garment is worn, shall be classified as a major defect.

2/ Any edge that is required to be turned under and is not, but is securely caught in the stitching, shall be classified as a raw edge. Any raw edge not securely caught in the stitching shall be classified as an open seam. Except on edge or raise stitching, a break in a line of stitching or two or more continuous skipped stitches or runoffs shall constitute an open seam. If either one or both sides of a double-stitched seam is open, it shall be classified as an open seam.

3/ An open seam includes seams with stitch tension such that stitch breakage occurs when strain is applied.

4.5 Examination of finished measurements. The undershirt and underpants shall be measured to determine conformance with 3.4.2 and 3.4.3. Each finished measurement deviating from the specified finished measurement shall be classified as a measurement defect. Sleeves or legs of the underwear uneven in length more than 1/2 inch shall be classified as a finished measurement defect.

4.6 Testing of materials and components. The finished basic fabric shall be tested as specified in table VI to determine conformance to the requirements specified in table I. The nylon filler yarns shall be tested for elongation as specified in 4.6.2.

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TABLE VI. Testing of basic fabric.

Characteristic (see table I)	Test Method
Total weight	ASTM D3776 (Option C)
Knitted cotton weight	ASTM D3776 (Option C)
Nylon filler weight	ASTM D3776 (Option C)
Thermal conductivity at 25 percent compression	ASTM C177
Thickness, uncompressed	ASTM D1777 1/
Thickness, compressed	ASTM D1777 1/
Elongation (finished fabric)	ASTM D5034
Air permeability	ASTM D737
1/ Woven and knitted materials, felts, nonwoven. A thickness gauge of the dead-weight type, equipped with a dial graduated to read directly to .001 inches, shall be used. The presser foot shall be circular, with a diameter of 1.129 +/- 0.001 inches and with moving parts weighted to a apply a total load of 0.60 +/- 0.03 pounds per square inch to the specimen. The anvil shall be not less than 1.129 inches in diameter. The presser foot and anvil surface shall be planed to within 0.0001 inches and shall be parallel to each other to within 0.0001 inches.	

4.6.1 Weight determination. The test sample shall be a piece of the finished material and a minimum of 4 square inches cleanly cut from the bulk of the material. No two samples shall be taken from areas containing the same wales and courses. Five samples shall be tested from each sample unit. An analytical balance capable of weighing to an accuracy of 0.01 gram shall be used. The samples shall be preconditioned in accordance with section 4 of FED-STD-191. The three layers of the sample (two layers of knitted cotton and one layer of nylon filler) shall be separated by cutting the tie stitches midway between the two cotton layers. After the layers have been separated, the cotton layers and the filler layer shall be weighed separately. The total weight of the two layers of cotton shall be considered as the weight of the knitted cotton for one sample and the weight of the filler shall be considered as the weight of the filler for one sample. The weight of the knitted cotton and the weight of the filler shall be added together to determine the total weight of the fabric in ounces per square yard. The weight of the sample unit, calculated as shown in equation (1), shall be the average of the results obtained from the samples tested.

$$(1)\quad \text{Weight } (oz/yd^2) = \frac{\text{Weight of sample}}{\text{Area of sample}}$$

4.6.2 Elongation (nylon yarns). A scale graduated in inches and fractions of an inch and of sufficient length to determine the maximum elongation shall be employed. A single yarn strand, minimum length of 3 inches and in a relaxed state, shall be placed along the length of the scale with one end of the strand at the end of the scale. The end of the strand at the end of the scale shall be held in a fixed position with one hand, and the opposite end of the strand shall be moved along the scale to the maximum extent that can be exerted by hand tension only. The length reached shall be recorded to the nearest 1/8 inch. The elongation percentage shall be calculated from the relaxed and the elongated dimensions. The average of the three tests results shall be considered the elongation percentage.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The underwear covered by this specification is intended for wear as the insulation medium with: Coveralls, Flying, Anti-Exposure CWN-21/P; Liner, Anti-Exposure, Flying Coveralls CWU-23/P; and Socks, Men's Anti-Exposure, Rubber SRU-25/P.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Size(s) required (see 1.2).

d. If a first article inspection is required (see 3.1).

e. Which Government-furnished patterns apply (see 3.5).

f. Whether a first article or conformance test report is required.

g. Packaging requirements (see 5.1).

6.3 Color shades. Samples of USAF color shades may be obtained from the procuring activity or as specified by the contracting officer.

6.6 Subject term (key word) listing.
Double knitted cotton
fasteners, snap type
Nylon yarn filler
Underwear, extreme cold weather
Underwear, two piece
Knitted cuffs

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

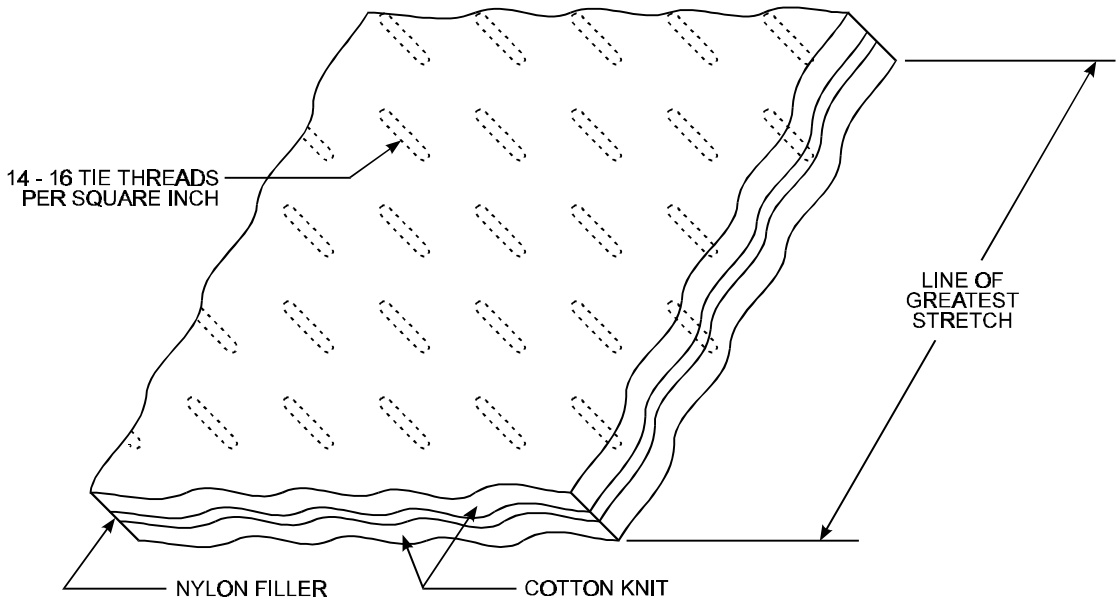


FIGURE 1. Basic fabric.

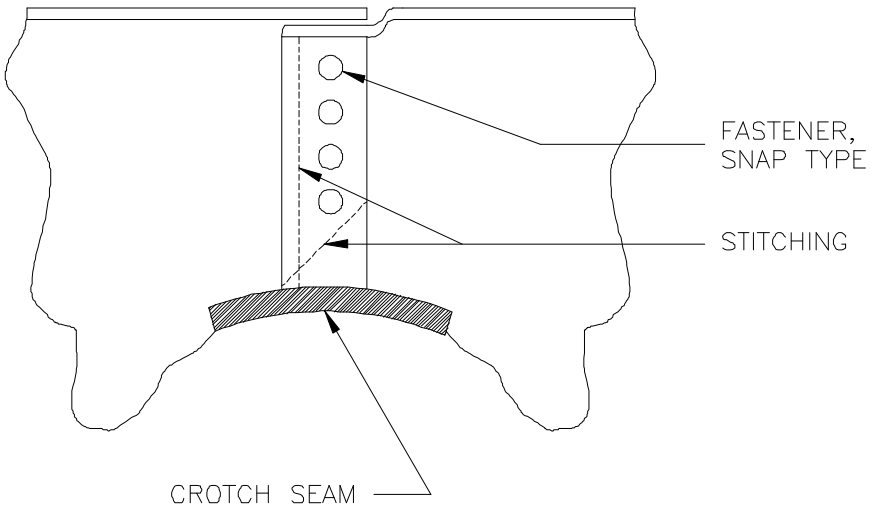


FIGURE 2. Fly stitching.

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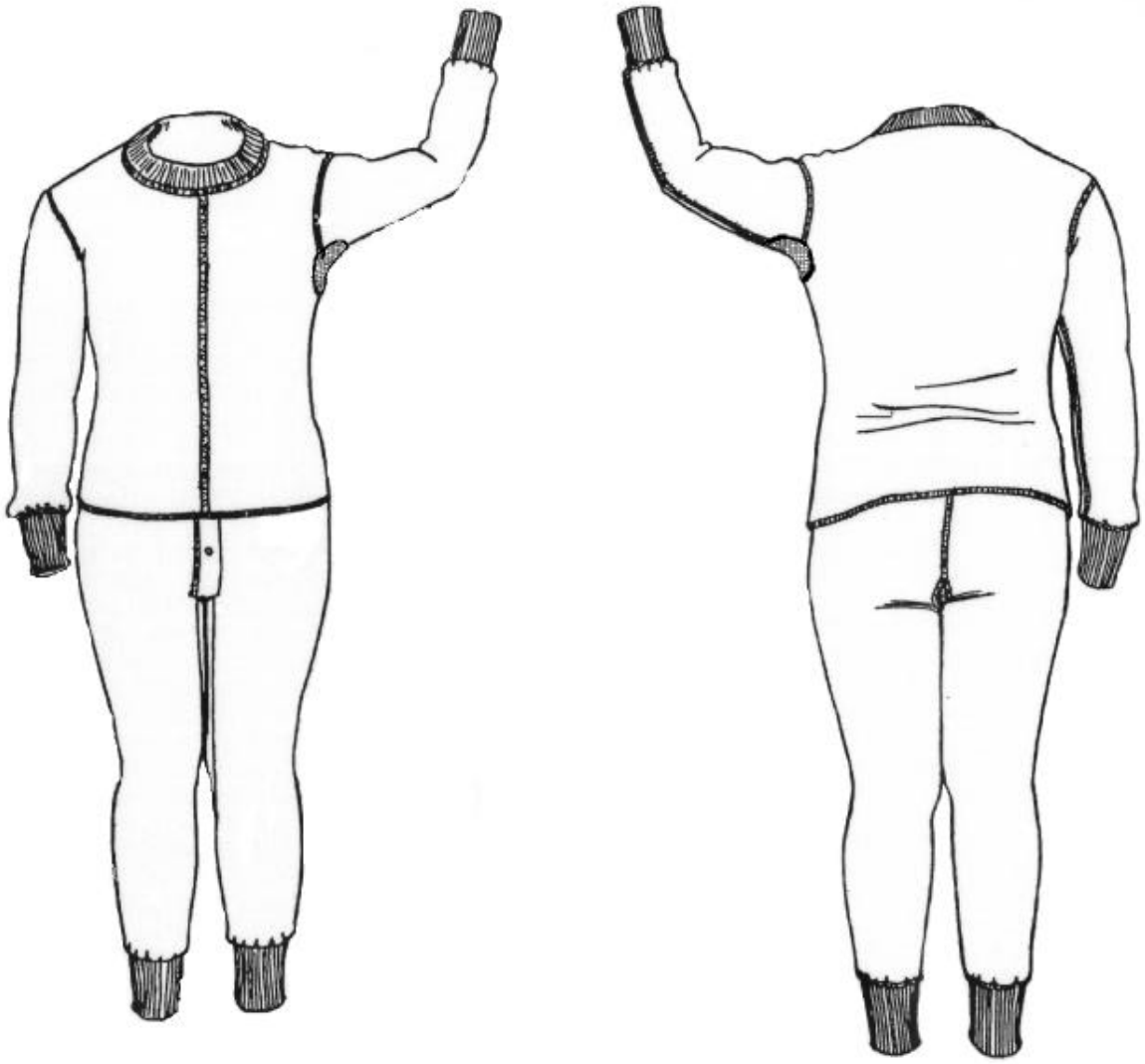


FIGURE 3. Underwear.

Custodians:
Air Force - 99

Preparing Activity:
Air Force - 82

(Project No. 8475-F003)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-83196C(USAF)	**2. DOCUMENT DATE (YYMMDD)** **98/11/30**

3. DOCUMENT TITLE
Underwear, Double-Knitted, Anti-Exposure, CWU-22/P

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. Organization	
c. ADDRESS *(Include zip code)*	**d. TELEPHONE** *(Include Area Code)* (1) Commercial (2) DSN *(if applicable)*	**7. DATE SUBMITTED (YYMMDD)**

8. PREPARING ACTIVITY

a. NAME SA-ALC/TILDD	**b. TELEPHONE** *(Include Area Code)* (1) Commercial 210-925-6314 (2) DSN 945-6314
c. ADDRESS ***(Include Zip Code)*** 485 Quentin Roosevelt Road Kelly AFB, Texas 78241-6425	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466** **Telephone (703) 756-2340 DSN 289-2340**

DD Form 1426, OCT 89 *Previous editions are obsolete.*

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